

MUHAMMAD RIZKY SANDYRA

(+62) 85155119405 | muhammadrizkysandrya26@gmail.com | [linkedin.com/in/mrizkysandrya](https://www.linkedin.com/in/mrizkysandrya) | <https://github.com/MRSandrya>

Padang, West Sumatera, Indonesia

SUMMARY

Fresh graduate in Informatics with hands-on experience building end-to-end machine learning systems, from a deep learning model (DenseNet-169) for medical image classification to an anomaly detection system for cybersecurity. Experienced in analyzing 2,400+ global income distributions and developing NLP pipelines for sentiment analysis using Python and Scikit-learn. Passionate about roles that combine data and artificial intelligence to deliver real-world solutions.

WORK EXPERIENCE

PT. Semen Padang

November 2025 - May 2026

Internship – Electrostatic Precipitator and Dust Collecting Maintenance Unit

- Developed a web-based Maintenance Service Order (MSO) application using the Laravel framework to digitize the maintenance process of ESP and dust collector equipment, improving tracking and work assignment efficiency
- Implemented CRUD operations, a multi-role authentication system (operator, technician, supervisor), and a performance report dashboard displaying key metrics such as MTTR, MTBF, and equipment status for maintenance data analysis
- Gained deeper understanding of industrial maintenance workflows, system integration with field requirements, and enhanced project management and debugging skills in production-scale software development

Dinas Komunikasi dan Informatika (DISKOMINFO) - Padang City

July 2024 - October 2024

Internship – Statistics and Cryptography Department

- Built a multi-vector attack detection system using Python to analyze server logs and detect 11 attack categories (SQL Injection, XSS, DoS, Brute Force, Path Traversal, etc.)
- Implemented regex pattern matching with a URL decoding pipeline to detect encoded attack payloads in Apache/Nginx log formats
- Developed behavioral anomaly detection based on IP profiling for automatic identification of Brute Force (>5 attempts) and DoS attacks (>100 requests)
- Automated the entire pipeline from raw log parsing, feature extraction, and classification to report generation using Pandas, producing structured CSV output for the security team's forensic analysis

EDUCATION

Universitas Negeri Padang

2021 - 2025

Bachelor of Informatics (S1)

- Focus: Data Analysis, Machine Learning, and Statistics
- Strong foundation in statistics, enabling meaningful insights from complex datasets
- Final Project: "Development of a Lung Disease Detection System Based on Deep Learning (DenseNet-169) for Medical Image Classification"

PROJECTS

Lung Disease Detection System — Deep Learning (DenseNet-169)

August 2025

- Built a multimodal lung disease classification system based on DenseNet-169 to detect 11 lung conditions from chest X-rays, achieving AUC 0.7586 (+111.5% above baseline)
- Implemented two-phase fine-tuning, CLAHE preprocessing, and integration of 13 clinical features on 458 X-ray images with custom Weighted Focal Loss to handle class imbalance up to a 15:1 ratio

Global Income Inequality Analysis (World Bank Data)

June 2025

- Analyzed 2,400+ country income distributions using World Bank dataset
- Calculated the Gini Coefficient and built Lorenz Curves to evaluate global income inequality
- Developed a simulation model to analyze the impact of redistribution policies (subsidies and progressive taxation) on changes in inequality levels
- Created data visualizations (time series, choropleth map, distribution curve) using Python

Sentiment Analysis on Tokopedia Product Reviews

June 2023

- Performed text preprocessing (tokenization, stopword removal, stemming) on a product review dataset
- Developed a sentiment classification model using machine learning (Scikit-learn)
- Evaluated model performance using accuracy and confusion matrix

COVID-19 Case Trend Analysis in South Korea

May 2023 - June 2023

- Conducted COVID-19 case trend analysis using a data-driven approach
- Used Python for data cleaning, exploratory data analysis (EDA), and trend visualization
- Identified case growth patterns as a basis for public health data interpretation

ACTIVITIES

Summer Course at INTI International University, Malaysia

May 2024

- Selected for an exclusive academic and cultural exchange program to broaden global perspectives
- Engaged in interdisciplinary learning to enhance professional and technology skills

MSIB Independent Study at MIKTI – Data Science

March 2024 - June 2024

- Participated in practical learning focused on data analysis and machine learning
- Applied data to support insight-driven decision making

INFINITE TECHNOLOGY UNP

2022 - 2024

CEO (2023)

- Led initiatives to drive innovation and improve digital literacy within the campus community
- Organized technology events that bridged industry gaps while developing leadership and strategic planning skills

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL
- **Data Analysis:** Pandas, NumPy, Data Cleaning, Exploratory Data Analysis (EDA)
- **Machine Learning:** Scikit-learn, Model Evaluation, Feature Engineering
- **Data Visualization:** Matplotlib, Seaborn, Tableau
- **Tools & Others:** Jupyter Notebook, Git, SQLite

OTHER

- **Certifications:** Huawei Machine Learning (HCIA-AI)
- **Languages:** Fluent in Indonesian and Conversational Proficiency in English